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A BRIEF HISTORY OF CHINESE MEDICINE IN SINGAPORE

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Singapore is a multiracial and multicultural country with ethnic Chinese (74%), Malays (13%) and Indians (9%) accounting for most of its population. It was a British colony from 1819 to 1963. During this time, it attracted waves of Chinese immigrants from China and around the Malay Peninsula from the nineteenth century onwards. Chinese medicine was brought into Singapore with the influx of Chinese immigrants. In the absence of a municipal system of medical care and poor relief, the Chinese medical delivery system comprising medical institutions, medical halls, clan-based recuperation centres and individual Chinese physicians, who operated from temples, market place or their own homes, filled a much-felt void in the lives of the Chinese plebeian classes (Yeoh 1991: 38–9). Chinese medicine remains relevant today. Singapore was recently praised by a World Health Organization (WHO) officer for pioneering good policies that protect against the perceived dangers of Traditional Chinese Medicine (TCM) resulting from poor integration with biomedical safety checks (Chan 2016).

Although Western medicine is the main form of healthcare in contemporary Singapore, Chinese medicine continues to enjoy considerable popularity. Chinese medicine practice in Singapore is confined to outpatient care. Chinese medicine practitioners are also called Chinese physicians, herbalists or therapists, but not doctors, as 'doctor' is reserved for registered Western medicine practitioners. According to a survey conducted by the Singapore government in 1994, 45% of Singaporeans had consulted a TCM practitioner at least once, and 19% of the population had visited TCM clinics in the last year. The proportion who had sought care from a TCM practitioner was highest among Chinese (54%), with 16% of Indians and 8% of Malays surveyed having also attended at least once. TCM was favoured slightly more by the elderly. The main reasons for consulting TCM practitioners were sprains, aches and pain, and the common cold (Ministry of Health 1995: 12).

There are some anthropological and sociological studies of health systems in Singapore, focussing on the phenomena of Chinese medicine practices and practitioners (Wu 1987: 71–94; Quah 1990: 122–59; Sinha 1996; Smith 2018). Historical research on Chinese medicine in Singapore is relatively scarce. Several Chinese physicians wrote about the important persons, institutions and events in the history of its development from the internal historical approach (Lee 1983, 1986; Tan 2001, 2007; Wong 2012). Some historians have emphasised the importance of Singapore's context in discussing the development of Chinese medicine

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by examining the linkages between the socio-political-cultural environments and the complicated modernisation process affecting Chinese medicine in Singapore (Foo 2000; Yang 2018). Although only limited literature has been published, there are copious amounts of primary sources available from various stages of Singapore's history. The sources include government records, reports and related documents about TCM in Singapore, annual reports and anniversary commemorative magazines of the Chinese medical organisations; academic journals published by these organisations, newspaper reports and op-eds on Chinese medicine, a series of oral history interviews conducted by the National Archive of Singapore with those key persons leading the development of Chinese medicine in Singapore, memoires and biographies of some influential Chinese physicians, manuscripts and collections passed down in some famous Chinese physicians' families.[1]

This chapter gives a brief introduction to the history of Chinese medicine in Singapore from the early colonial period to the present day. The discussion is divided into three sections according to the three different agents acting towards the development of Chinese medicine in Singapore. The three agents are governments of different time periods, Chinese medical professional associations and religious organisations. By showcasing how these agents have led and influenced the development of Chinese medicine (institutionally and practically), this article highlights the unique characteristics and the circuitous modernisation process of Singapore's Chinese medicine.

State laws and policies

The Medical Registration Ordinance of 1905, which was the first medical law under the British administration, set a framework for the control of medicine in Singapore. Article 21 of this ordinance states:

> Nothing contained in this Ordinance shall be construed to prohibit or prevent practice of native systems of therapeutics according to Indian, Chinese or other Asian Method.
>
> *(Straits Settlements 1920: 276)*

This law has had a profound and long-lasting impact on the control of Chinese medicine in Singapore. It has three implications. First, it guaranteed that the government would not prevent or prohibit TCM practices, leaving a space for Chinese medicine to prosper. Second, it showed that the only official healthcare system is based on Western medicine, not Asian medicine which is portrayed as 'native' and traditional, in contrast to the image of Western medicine as a modern, translocal science. Third, it forbad the integration of Western medicine and Asian medicine, unlike current practice in China today. It set a clear boundary to separate the two different medical systems, one which has lasted until now. Due to the exclusion posed by the law, Chinese medical practices were considered as local businesses, so that Chinese physicians had to pay for business registration fees and signboard taxes.

During the Japanese occupation of Singapore (1942–1945), all Chinese physicians were requested to obtain a certificate of registration under the Japanese administration. The new government announced that all doctors, dentists and native doctors, including practitioners of Chinese, Malay and Indian medicine, must register before they start their profession. This was the first time that Chinese medicine was directly controlled under the government. Chinese physicians had to attend exams assessing their abilities. Only those who passed could

register and were given quotas of Chinese herbs. The purpose of this was to control the herb supply strictly and hence to tighten control over the Chinese community.

After the war, the British returned and started the Medical Plan expanding healthcare services to the whole local population on the island. This marked the beginning of the dominant role of Western medicine, which had lasting effects on Singapore's current healthcare system until now. While modernising its medical system, the Singapore government also initiated the indirect control of Chinese medicine through the Medicine (Advertisement and Sale) Act 1955. Since Chinese medical practices were considered business undertakings, the government restricted the selling and advertising of all medicine, Chinese medicine in particular. The law prohibited exaggerated and unscientific advertisements of Chinese medicine by banning key words such as 'guaranteed to cure' in reference to certain diseases and conditions including blindness, cancer, cataract, dangerous drug addiction, deafness, diabetes, epilepsy or fits, hypertension, insanity, kidney diseases, leprosy, menstrual disorders, paralysis, tuberculosis, sexual function, infertility, impotency, frigidity, conception and pregnancy (Singapore Government Printing Office 1955: 1–3). There were lots of such advertisements for Chinese medicine in Chinese newspapers. The target of the law was obvious. Unfortunately, the law was not effectively enforced as Chinese advertisers found ways to change the key words without changing the meaning. For instance, they used 'waist (*yao* 腰)' to replace 'kidney (*shen* 肾)' since *shen* 肾 was explicitly prohibited in the law, but kidneys are well-known to govern the waist and lower back in Chinese medical theory. Nevertheless, this indirect control of Chinese medicine, together with the underlying logic to erase unscientific elaboration and to emphasise the importance of science is interesting to note in the light of later changes.

In 1965, the Republic of Singapore was established. One of the colonial legacies was the laws on medicine and related policies were retained. Chinese medicine was still not under the direct legal control of the government. In order to gain an understanding of Chinese medicine and to introduce direct control, the government invited doctors to conduct scientific comparative research on Chinese medicine with the cooperation of Chinese physicians (Gwee *et al.* 1969). The aim of the study was to see if there was any correlation between the two systems of medicine in the way of approach and diagnosis, and to discover the similarities and the differences between the two medicines. The roots of undertaking these investigations within a modern scientific framework can be traced to colonial period medical governance. The rationale behind this action also explains later measures taken by the government. The findings of the study emphasised the incommensurability between Chinese and scientific medicine, a difference which informed later conflicts, even though the study also found some common elements across the two diagnostic methods.

This epistemological difference came to a head in the prohibition on the use of berberine (*huangliansu* 黄连素), implemented under the Poisons Act in 1978. The drug advisory committee, which was made up of doctors and appointed by the government, announced that *huanglian* was to be banned because Western medical tests had proved it harmful to babies and pregnant women. The Singapore Chinese Physicians' Association (SCPA) appealed against the ban, writing to the minister for health and through newspaper editorials, claiming that *huanglian* was an important and useful drug in Chinese medicine (Singapore Chinese Physicians' Association 1995: 767). The committee insisted on doing scientifically controlled trials and finding statistically significant results to show the importance of *huanglian*. The SCPA listed out research papers by Chinese physicians in China as contradictory proof.

These papers were not accepted by the committee as they were not regarded as scientific enough because they were grounded in TCM theory. This showed that the government's stance on Chinese medicine was to allow the practice of Chinese medicine, but only on the basis of Western scientific criteria. This case was the first time when biomedical argumentation about scientific method was relied on in a case of law to regulate Chinese medicine. This was the beginning of using biomedical standards to control and regulate Chinese medicine in an open manner.

The Traditional Chinese Medicine Practitioners Act was enacted in 2000, marking a transition from regulating Chinese medicine through the Poisons Act, as in the case of berberine above, to the start of state regulation of TCM as a whole. The framework of this act is exactly the same as the acts on Western medicine, including the establishment of a Traditional Chinese Medicine Practitioners Board (TCMPB). Thus, the governance of TCM was copied directly from the governance structure of Western medicine. The role of the TCMPB is similar to the Singapore Medical Council, but targeting Chinese physicians instead of doctors. The ethical code and guidelines for Chinese physicians were also modified from the code and guidelines for biomedical doctors. The investigation procedures and penalties for misconduct by physicians are modelled on regulations for doctors. Continuing past practice, TCM practitioners are not allowed to use Western diagnostic equipment or dispense Western pharmaceuticals. Chinese Proprietary Medicines must not be adulterated with Western drugs.

In addition, the government implemented a new framework for administering clinical practice, teaching and research on Chinese medicine, based on the existing biomedical model, in order to complete a top-down modernisation of Chinese medicine. Here, further discrepancies arose due to the differentials of epistemology and institutional power between the two medicines. For example, the registration of Chinese physicians was different from the registration of doctors. Since the situation was complicated, the government learnt from the experience of Hong Kong and took a phased approach by invoking the grandfather clause during the transitional period. Those considered as older practitioners were grouped into four categories according to the years of experience and the certificates they had obtained. Some were fully or partially exempted from taking the standardised qualifying exam.

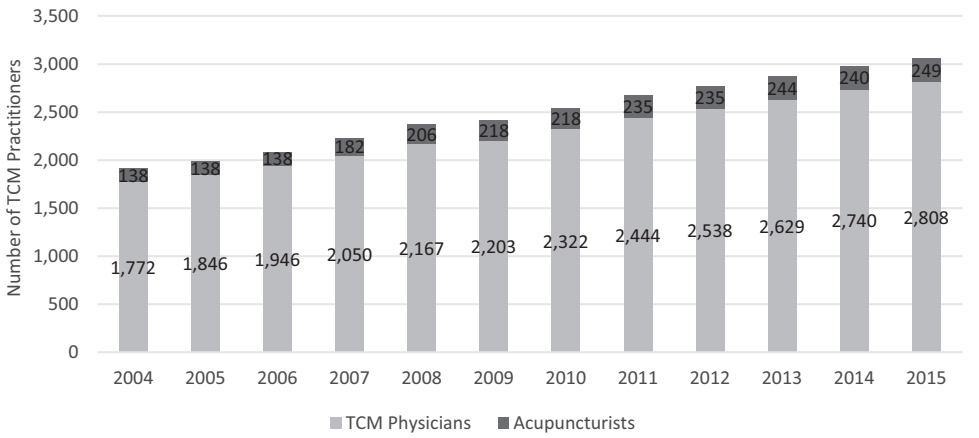


Figure 36.1 Number of TCM Practitioners on the Register, 2004–2015

From 2004 onwards, the transitional period was over. All physicians have to pass the national qualifying exam to get registered (Figure 36.1).

The training approved by the government duplicated the training system in Singapore medical schools. In the training aspect, the government copied the system for Western medicine doctors: introducing the new undergraduate double degree programme and continuing education programmes for Chinese physicians to upgrade their qualifications. The five-year double degree programme in Biomedical Sciences and Chinese Medicine is jointly organised by Nanyang Technological University in Singapore and Beijing University of Chinese Medicine in China. Within this framework, the curricula and the degrees offered by the university are subject to approval by the Ministry of Education. The continuing education programmes are organised mainly by professional associations with permission from the Ministry of Health.

In the research aspect, government encourages collaboration between Chinese physicians and Western medicine doctors and scientists in carrying out clinical research to provide scientific evidence. More and more TCM research grants were given to support research using scientific methodology.[2] All of these measures show the government's emphasis on biomedicine and the dominance of biomedical system.

Apart from the laws, the policies related to Chinese medical institutions show that the government is keen to promote Chinese medicine because of its potential to alleviate government healthcare cost. This support for Chinese medicine can be traced to the colonial period. Unlike in Hong Kong, Chinese medicine practices were endorsed and supported by the colonial government via land grant and tax exemption for TCM charity clinics. Although there were lots of complaints and criticisms from Western medical officers, the colonial government found that there was a need to allow Chinese medicine for healing the Chinese due to the scarcity of medical resources (Mugliston 1893/1894: 72). Thong Chai Medical Institution (*Tongji yiyuan* 同济医院), founded in 1867, was the earliest charitable organisation to provide free Chinese medical treatment and free Chinese herbs to help the needy sick. The Colonial Secretary of the Straits Settlements praised Thong Chai for its charitable work and granted a piece of land for Thong Chai to build its own premise. Other TCM charitable institutions, including the Chung Hwa Medical Institution (*Zhonghua yiyuan* 中华医院), enjoyed other benefits, such as exemption from paying property tax or licence fees.

These benefits continued after Singapore became an independent republic. Citing the ageing population and increasing burden of healthcare cost as their reason, the government continued providing incentives to support TCM charity clinics (Thong Chai Medical Institution 1979: 22–3). These expanded rapidly, which was in line with the government's community development masterplan. In each district, there is an average of six charitable TCM clinics providing free or cheap TCM service for the community, making it convenient for the elderly who live nearby. It is important to note that the tradition of charity practised by these Chinese medical institutions encourages Chinese medicine to flourish in Singapore.

The roles of professional associations

Professional associations have played a key role in developing Chinese medicine in Singapore since the government promoted self-regulation by the TCM community before implementing statutory regulation. There are eight self-funded non-governmental professional associations in Singapore (Table 36.1). Thong Chai is an exception as it takes the form of a medical institution rather than a professional association, yet it plays a role similar to the associations in building the training and research system. These grassroots efforts have been extremely

Table 36.1 TCM Associations in Singapore

Association	*History*	*Affiliated Organisations*	*Members*
Singapore Chinese Medical Union (*Zhongyi zhongyao lianhehui* 中医中药联合会)	Founded in 1929 under the leadership of Li Bo Gai 黎伯概and Eu Yan Sang Medical Hall 余仁生药行, suspended in 1942, resumed in 1947. The current president is Guo Jun Xiang 郭俊缃.	TCM Cultural College (*Zhongyiyao wenhua xueyuan* 中医药文化学院)	Around 200 members in 1954, 141 members in 1989, around 200 members in the 2000s.
Singapore Chinese Physicians' Association (the SCPA, *Zhongyishi gonghui* 中医师公会)	Founded in 1946 under the leadership of Wu Rui Fu 吴瑞甫, You Xing Nan 游杏南 and Zeng Zhi Yuan 曾志远.	Chung Hwa Medical Institution (previous name as Chung Hwa Free Clinic), Singapore College of Traditional Chinese Medicine (*Xinjiapo zhongyi xueyuan* 新加坡中医学院, previous name as Singapore Chinese Physicians' Training College), Chung Hwa Medical Research Institute (*Zhonghua yiyao yanjiuyuan* 中华医药研究院), Chung Hwa Acupuncture Research Institute (*Zhonghua zhenjiu yanjiuyuan* 中华针灸研究院)	Largest organisation by number of members, around 600 members in 1980, 879 members in 1990, 1200 members in 1994, 1169 members in 2006, 1600 members in 2018.
Association for Promoting Chinese Medicine (*Zhongyiyao cujinhui* 中医药促进会)	Founded in 1957 under the leadership of Fang Zhan Lun方展纶, previous name was *Zhongyiyao tilianyaoye cujinhui* 中医药提炼药业促进会, Re-established in 1971 under the leadership of Chen Zhan Wei 陈占伟.	Public Free Clinic Society (*Dazhong yiyuan* 大众医院), Institute of Chinese Medical Studies (*Zhongyixue yanjiuyuan* 中医学研究院), Research Institute of Chinese Medicine and Acupuncture (*Zhongyiyao zhenjiu yanjiuyuan* 中医药针灸研究院)	500 members in 2007.
Singapore Acupuncture Association (*Zhenjiu yixue xiehui* 针灸医学协会)	Founded in 1977 under the leadership of five doctors and five acupuncturists including Zhang Yu Hua 张玉华, renamed as *Xinjiapo zhenjiu xuehui* 新加坡针灸学会 in 1998.	School of Acupuncture (*Zhenjiu xueyuan* 针灸学院), Acupuncture Charitable Center (*Xinjiapo zhenjiu shizhenzhongxin* 新加坡针灸施诊中心)	85 members in 1995, 87 members in 2018.

(*Continued*)

Association	*History*	*Affiliated Organisations*	*Members*
Society of Chinese Medical Research (*Xinyi xiehui* 新医协会)	Founded in 1984 under the leadership of Cai Yu Quan 蔡玉泉.	Kang Ming Chinese Physicians College (*Kangmin yiyao zhenjiu xueyuan* 康民医药针灸学院), Kang Ming Clinic (*Kangmin yiyuan* 康民医院)	Small organisation as the members are from the graduates of Kang Ming college only. Number of members is not mentioned.
Xinhua Medical Society (*Xinhua zhongyiyao xiehui* 新华中医药协会)	Founded in 1994 under the leadership of Chen Pan Xu 陈磐绪, merged into Singapore Chinese Medical Union in 2000.	Jian Min Medical Hall (*Jianmin yaofang* 健民药房), Jian Min College of Chinese medicine (*Jianmin zhongyi xueyuan* 健民中医学院)	Small organisation as the members are from the graduates of Jian Min college only. Number of members is not mentioned.
Singapore Chinese Physicians' Training College Alumni Association (*Zhongyi xueyuan biyeyishi xiehui* 中医学院毕业医师协会)	Founded in 1958 under the leadership of student council in Singapore Chinese Physicians' Training College, renamed as Singapore College of TCM Alumni Association in 2012.	Singapore College of Traditional Chinese Medicine	Around 80 members in 1959, 282 members in 2017.
Society of Traditional Chinese Medicine (*Xinjiapo zhonghua yixuehui* 新加坡中华医学会)	Founded in 2000 under the leadership of Lee Kim Leong 李金龙	Specialist Traditional Chinese Medicine Center (*Zhuanke zhongyiyuan* 专科中医院), Academy of Traditional Chinese Medicine (*Zhuanke zhongyi yanjiuyuan* 专科中医研究院)	Around 200 members in 2018, the member must possess a master or doctorate degree in Chinese Medicine from TCM universities in China.

important in the institutionalisation and modernisation of Chinese medicine in Singapore, not only regulating the profession, but also leading the teaching and research of Chinese medicine without governmental guidance prior to 2000.

As shown in the table, these professional associations have a similar organisational structure, which is a three-in-one or four-in-one integrated structure. Each association is like a conglomerate assembling three or four subsidiaries under one umbrella. These subsidiaries, including clinics, schools and colleges and research institutes, focus on the areas of clinical practice, teaching and research. These umbrella-shaped institutions form a kind of guild which regulates the physicians. The highlight of this integrated structure is the interconnections among these entities that allow a better integration of different important aspects (clinical practice, education and research), essential to the modernisation of TCM. This organisational structure is also self-sufficient as it consists of all the essential aspects which provide support and sources to each other. Because of this unique structure, these associations played the key role of leading the development of Chinese medicine without help from the state before 2000.

Taking the SCPA as an example, the SCPA started its role as a gatekeeper and regulator in 1947, fulfilling a vacuum left by the colonial government. The SCPA announced that it followed the Physicians Act (*Yishifa* 医师法) of the Republic of China (ROC) promulgated in 1943. The SCPA considered that Chinese physicians overseas were still under the control of the ROC as they were Overseas Chinese. According to the act, SCPA was supposed to be the only legitimate local professional association in the region. The association started the work of registering physicians and regulating the profession according to the act. After Singapore gained independence, this situation changed completely as these physicians were no longer considered Chinese, but Singaporean/Malaysian physicians, and ROC laws no longer applied to them. Therefore, the SCPA started drafting its own principles and regulations. It referred to the previous Physicians Act and formulated new rules according to the local situation. These rules and regulations, including registration requirements, practice guidelines and disciplinary procedures, were revised several times to fit in with the local context and to accommodate the needs of different periods (Yang 2018: 128–49). For instance, the registration of physicians with the SCPA was initially open only to graduates from the SCPA-affiliated training college. Outsiders who wished to join the association had to be recommended by SCPA members and pass an exam in order to register. When the government called for the formation of one unified body to oversee all physicians, an initiative that lasted from 1980 to 1992, the association changed its entry requirement to recruit other physicians who had other certificates or equivalent experience without exam or member recommendation. The government ultimately gave up the endeavour, due to ongoing internal conflicts between various professional associations.

Membership in the SCPA was a symbol of professionalism. Being registered with the SCPA meant the physician was a qualified Chinese medicine practitioner certified by the SCPA and held the licence issued by the SCPA. The physician had to abide by the rules and regulations of the SCPA. Offenders would be investigated and punished according to society rules. The SCPA also introduced new measures such as issuing medical certificates to its members, an approach they learnt from the Western medical system. These efforts at self-regulation were effective, bringing a normalisation of practice and excluding poor practitioners to some extent, and had a significant impact on the Singapore community before 2000, when the Ministry of Health stepped in.

Besides these regulatory practices, the SCPA also established broad and extensive networks with overseas organisations, including TCM universities, international associations

and government bodies in China. These networks have been helpful in raising TCM standards in Singapore, as they facilitate the exchange of resources and promote the interaction and collaboration between organisations. Members of the SCPA enjoy the privileges of attending workshops, and going for exchanges and internships to partner institutions. The SCPA also invites distinguished foreign experts to deliver lectures and conduct courses for its members.

Under the SCPA umbrella, the clinics, colleges and research institutes affiliated to the SCPA share resources and take advantage of collective international networks. The principal and teachers in the colleges are specialists or senior physicians who also work in associated clinics. They are also leaders of research groups in Chung Hwa Medical Research Institute and Chung Hwa Acupuncture Research Institute. Some of them also sit on the council of the SCPA. Students of the college are sent to SCPA clinics to observe and for internships. They have to participate in research groups in research institutes, where the projects are based on patient records and therapeutic practices used in the clinics. After graduation, they become members of the SCPA, and can continue to participate in activities hosted by affiliated organisations. The facilities and resources, such as the library and the labs, are also shared among these institutions.

This self-sufficient four-in-one structure is only possible because of the philanthropy of the physicians and the public. It is important to note that the school teachers, the researchers, the physicians working in the clinics and the council members in the SCPA are all volunteers. The clinics affiliated to the SCPA are all charitable organisations which rely on generous donations from the public. The clinics provide free resources for training and research.

The college affiliated to the SCPA is the largest TCM training school in Singapore. The history of this college shows the transformation of the training system of Chinese medicine from the traditional diverse acquisition methods to the modern standard college system. The college was founded in 1953. Because of restrictions imposed by the immigration ordinance issued by British colonial government in 1952, fewer physicians were allowed to enter Singapore. Facing a lack of successors, the SCPA initiated a plan to set up a TCM training school. Before this, there were different traditional acquisition methods such as apprenticeship training and self-learning. At the beginning stage when the school just opened, the key figures in the SCPA were the teachers. They wrote their own handouts and compiled their own textbooks. Their way of teaching was more like a master-apprentice model as there was no standard syllabus or curriculum. The situation changed in the 1960s when the school started to adopt the standard textbooks from China. Referring to the courses and curricula offered in TCM colleges in China, the teachers made some adaptations to meet local students' needs and designed a similar curriculum which was more suited to the Singaporean context.

Enrolment requirements also changed over the years. In the beginning, there were no specific degree requirements. From 1980 onwards, the entry requirement was changed to allow only those who had completed their junior college education. Those who did not meet this requirement had to take extra bridging courses before enrolment. Compared to China, students in Singapore's TCM colleges are from quite diverse social backgrounds. Most of them are middle aged and older, from different industries and income groups. About half of the students quit half-way through their training, and only 50% of graduates go on to pursue a professional career. The others study for personal benefit, such as learning self-care. This should be viewed in the light of the fact that TCM was not regulated as a discrete branch of medicine under the national healthcare system until 2000.

The main differences between the training in Singapore and China are,

1 Singaporean colleges only offer part-time courses;
2 clinical practice usually starts from year one onwards; [3]
3 across similar subjects (such as *Inner Canon of the Yellow Emperor*, Warm Diseases 温病, Injury from Cold 伤寒, etc.), the instruction takes more class hours in Singapore as theoretical teaching is integrated with clinical practice. This is in line with the college mission to train TCM professionals with an emphasis on clinical capabilities and experience;
4 the focus on biomedicine is much less than in China as Singapore forbids the integration of Chinese medicine and Western medicine. Thus, colleges teach very few biomedical or science subjects.

After China opened up in the 1980s, China had a greater and more direct influence on TCM education in Singapore, while more China experts were invited to give lectures and to conduct training courses. More and more Singaporean physicians went to China to undertake postgraduate training. From 1994 onwards, local colleges started joint degree programmes with renowned TCM universities in China. These joint degree programmes were considered to have high standards of teaching and gained public recognition. This recognition was important as the government had yet to assess the standards of training since it had not legislated concerning TCM.

From the aspect of research, Singaporean professional associations have carried out research studies of TCM from the beginning to now. Before the 1950s, physicians tried multiple diverse research methods and methodologies. These included not only traditional approaches, such as textual research and annotation of medical classics, but also different scientific methods, including basic chemistry, keeping statistical results, systematic observation and logical deduction. Singapore has been creating a new modern industrial society since the 1950s, in which the government emphasises the study of mathematics and science. English is used as the medium of instruction, which also helps to spread Western influence. As a result, this environment has stimulated scientific research into TCM. A major hindrance faced by physicians in Singapore is the prohibition of integration of Chinese medicine and biomedicine. Because of this, researchers can only use the mathematical and scientific approaches from physics, chemistry and biology to do analysis and to conduct experiments. Researchers also focus more on the dialectical methods of pattern differentiation and treatment determination (*bianzheng lunzhi* 辨證論治) and traditional theories of TCM, which are different from the Westernised approach adopted by some TCM researchers in China.

The research institutes affiliated to the professional associations organise workshops and symposia, publish academic journals and set up specialty groups to promote academic research. Local physicians usually collaborate with experts in China who are their teachers or classmates. They consult these experts for ideas and theories while carrying out clinical research in Singapore on the local population. Sometimes they ask for technical support from overseas institutions due to the lack of facilities, as these research institutes are self-funded and small-scale.

Overall, these professional associations have made invaluable contributions to the development of TCM in Singapore, in such areas as the self-regulation and licencing of TCM practice, training of TCM practitioners and leading research on TCM (Yang 2018: 101–273).[4] They initiated a bottom-up modernisation by selectively learning from China's

experience and adapting to fit into Singapore's context, implementing measures, such as forbidding the integration of Chinese and Western medicine, and assisting in the development of government-directed training programmes. The complicated process of development shows indigenous innovation and continuous adaptation. After the legislation on Chinese medicine, the government and the professional associations, via negotiation and collaboration, continued to modernise Chinese medicine.

As a response to the government's request, representatives from the first seven of the associations listed in Table 36.1 and Thong Chai formed the Singapore Traditional Chinese Medicine Organisations Coordinating Committee (*Zhongyi tuanti xietiao weiyuanhui* 中医团体协调委员会) in 1995. This committee, representing the TCM community, played a key role in negotiations with the government in the transition phase. It drafted the 'Guiding Principles Memorandum for Singapore Chinese Medical Education' and 'Practice Guidelines and Ethical Codes for Singapore TCM Practitioners'. It also certified the physicians and published the 'List of TCM Practitioners in Singapore'. Their work was based on their previous experience and provided important input for the government.

The actions taken by religious organisations

Apart from scholarly medicine, folk medicine (Topley 2011: 451–3) has coexisted in Singapore until now. Before the mid-twentieth century, some physicians even used Daoist rituals to treat patients or included Daoist philosophical mores in their prescriptions. This phenomenon started disappearing from the 1930s onwards after the physicians began to draw a strict line between scholarly and folk medicine and tried to eradicate what they saw as mere superstition.

Nevertheless, folk ideas and methods have been preserved and practised by religious professionals in temples even until now. We can still observe the combination of ritual and medical treatments in such places of worship as the Temple of Baosheng Emperor (*Zhenren gong* 真人宫), Singapore Teochew Charity Hall (*Xiude shantang yangxinshe* 修德善堂养心社) and Kiew Lee Tong Temple (*Jiuli dong* 九鲤洞). The common feature of these temples is that the Gods of Chinese medicine such as Hua Tuo 华佗, Wu Tao 吴夲 or Ji Gong 济公 are worshipped in these temples. Patients who seek treatment at these temples are considered to be seeking treatment from divine physicians. The seeking of treatment at these temples is understood to be by patients seeking help from the divine physicians. The types of healing include asking for prescription divinations (*Yaoqian* 药签) and getting prescriptions through the writing of spirit-diviners (*Fuji* 扶乩) after communication with the deity.

Over the years, there have been two prominent trends in the development of folk medicine. The religious organisations which provide these treatments started taking two steps. One is to re-explain and validate the rituals/activities by looking for scholarly proof. The other is to develop and institutionalise secular medicine by setting up and expanding affiliated free TCM clinics. Some physicians in these clinics may promote TCM knowledge as an expression of their religious ethics. These changes have both been influenced by the rationalisation of the country.

Conclusion

The Singapore case offers an interesting example of how Chinese medicine has been developed in an overseas context, affording a view of how Chinese medicine has adapted to conform to local conditions. Due to both internal and external influences, Chinese medicine

has undergone a unique modernisation which is different from TCM in China. The non-governmental organisations, including professional associations and religious groups, were instigators of the professionalisation from below. The government, via collaboration with the professional associations, completed the top-down modernisation. The development of Chinese medicine in Singapore is a metaphor of Singapore, illustrating a blend of East and West embodying the interaction of Chinese influence and the British colonial legacy.

Notes

1 For a full list of these primary sources, refer to Yang Yan (2018: 350–61).

2 For the list of research projects funded by the government, refer to https://www.moh.gov.sg/docs/librariesprovider5/research-grants/application-guide-for-tcmrg-(jan-2017).pdf, accessed 8/10/2018. For information on how government funded research projects and how their terms and conditions follow a biomedical frame, see Yang (2018: 93–4). Two examples will suffice here to show the predominance of the biomedical method in government-funded research:

> K.H. Kong (2018) 'Acupuncture in the Treatment of Fatigue in Parkinson's Disease: A Pilot, Randomized, Controlled, Study' and Singapore General Hospital (2017) 'Traditional Chinese Medicine for Treatment of Irritable Bowel Syndrome'.

3 Colleges started to offer full-time courses only after the legislation of TCM under the supervision of TCMPB.

4 One example of new research is the innovative new proprietary Chinese medicine, *Bushen Yijing Wan* 补肾益精丸. Thong Chai Medical Research Institute invented this pill and successfully got the patent. For the research paper related to this development, see Lu (2015).

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